



Applications of E-nose, GC-MS, and GC-IMS in tea volatile components analysis

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ABSTRACT

This study reviews applications of Electronic Nose (E-nose), Chromatography-Mass Spectrometry (GC-MS), and Gas Chromatography-Ion Mobility Spectrometry (GC-IMS) in tea industry research. The E-nose offers rapid, non-destructive analysis for real-time quality assessment through odor fingerprint pattern recognition, yet cannot identify or quantify individual compounds. Gas GC-MS, established as the industry gold standard, enables precise qualitative/quantitative analysis of complex volatiles with enhanced efficacy for higher molecular weight compounds. GC-IMS emerges as a user-friendly tool for highly sensitive detection of low molecular weight volatile compounds and rapid two-dimensional gas separation, effectively compensating for GC-MS limitations in trace-level small molecule analysis; however, its compound identification accuracy remains inferior to GC-MS due to incomplete spectral libraries. Collectively, these technologies enable comprehensive tea quality evaluation by characterizing aroma profiles across varieties, authenticating geographical origins, and tracking volatile changes during processing and storage. Their complementary strengths in speed (E-nose), specificity for macromolecules (GC-MS), and sensitivity to small molecules (GC-IMS) establish a multidimensional framework for tea research. This integration provides robust solutions for classification, process optimization, and shelf-life studies, offering theoretical and practical insights to advance quality control, product development, and origin traceability in tea production chains.

1. Introduction

Tea, one of the three most consumed non-alcoholic beverages globally, ranks second only to water in global consumption. It is renowned not only for its distinct flavor and nutritional value but also for its profound historical and cultural heritage. Originating in China, tea's discovery dates back to 2737 BC, as documented in the Shennong Ben Cao Jing (Classic of Herbal Medicine), which attributes its discovery to the legendary Emperor Shennong. Tea cultivation and consumption began during the Shang Dynasty (1500–1046 BC), marking the inception of tea utilization (Samanta, 2022). Tea is divided into six categories, in order of fermentation degree: green tea, white tea, yellow tea, oolong tea, black tea, and dark tea. Fig. 1 illustrates their respective production processes, while Table 1 summarizes the characteristic features of each tea category. The unique aroma of tea arises from its volatile organic compounds (VOCs). High-quality teas are typically characterized by superior aroma and taste, while lower-quality teas exhibit inferior sensory attributes in these aspects. Consequently, a desirable aroma is

critically significant in tea quality evaluation, serving as a key determinant in comprehensive quality assessment systems. It is intrinsically linked to the overall quality characteristics of tea and profoundly influences consumer sensory experience, market acceptance, and competitiveness (Banerjee, Tudu, Shaw, Jana, Bhattacharyya, & Bandyopadhyay, 2012).

The electronic nose (E-nose), a biomimetic simulation of the human olfactory system, has emerged as an efficient alternative to traditional methods for analyzing food flavors and odors. Functioning as an intelligent sensing device, the E-nose's core architecture comprises a sensing system and an automated pattern recognition system. The sensing system, which may adopt configurations such as a multi-sensor element array, a single device, or a hybrid of both, deploys gas sensor arrays with overlapping selectivity to convert reaction changes induced by target gases into detectable electronic signals. These signals manifest as variations in parameters such as resistance, current, voltage, absorbance, or vibration frequency, forming the data foundation for subsequent pattern recognition (Berna, 2010; Lim, Swamy, Ramakrishnan, Chan, &

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Kesuma, 2023). Leveraging its high-sensitivity detection capabilities, E-nose technology has been widely applied in trace detection and real-time monitoring of VOCs across diverse fields, including food quality monitoring, environmental pollutant detection, medical diagnostics, and energy exploration.

Gas chromatography-mass spectrometry (GC-MS) is a highly versatile analytical technique ideally suited for comprehensive analysis, as it combines high separation efficiency with multifunctionality, selectivity, and sensitive mass detection (Koek, Jellema, van der Greef, Tas, & Hankemeier, 2011). The system comprises two core components: the gas chromatography unit, which separates complex mixtures, and the mass spectrometry unit, which identifies unknown compounds through manual spectral interpretation or computer-assisted spectral library matching (Gould, Nguyen, & Honeychurch, 2023). Leveraging its multidimensional data analysis capabilities, GC-MS provides precise and reliable analytical solutions for fields such as environmental monitoring, pharmaceutical research, and food quality testing. With ongoing advancements in hyphenated technologies, future GC-MS systems are

expected to achieve breakthroughs in intelligent automation and miniaturization, further broadening their application scope.

Gas Chromatography-Ion Mobility Spectrometry (GC-IMS) is an emerging analytical technology that couples gas chromatography with ion mobility spectrometry, enabling fingerprint analysis of VOCs in samples without requiring complex pretreatment (L. C. Zhang, Shi, Sun, Xie, Wang, Shi, et al., 2024). This technique is characterized by rapid response, high sensitivity, and excellent stability. Currently, it has been widely applied in odor detection, quality assessment, and product classification of diverse volatile substances, including foods, teas, and traditional Chinese medicines (W. Q. Li, Chen, Blank, Li, Li, & Liu, 2021; M. Wang, Li, Ding, Chen, Liu, Wang, et al., 2023; Yang et al., 2022). In the field of odor analysis, GC-IMS is increasingly recognized as a versatile and practical tool for VOC detection, akin to a "Swiss Army knife" in analytical science. It effectively addresses complex detection challenges by delivering precise and comprehensive results, thereby advancing quality control and characteristic research in related industries (Hadi & Philipp, 2024).

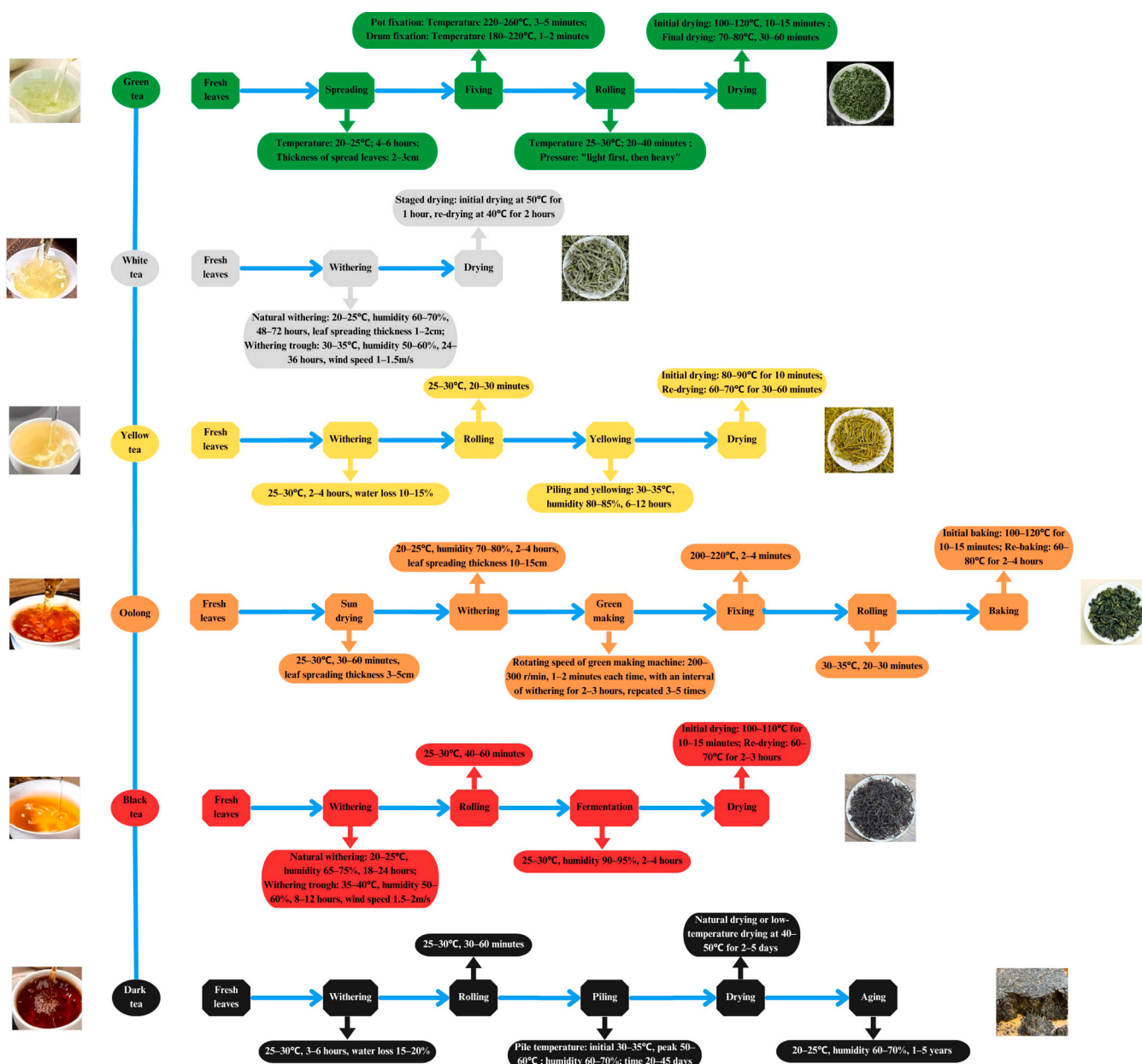


Fig. 1. Comparison diagram of the entire process of six major tea processing technologies.

This study focuses on research literature integrating tea with E-nose, GC-MS, and GC-IMS, systematically reviewing and thoroughly exploring the current applications and future trends of these three detection technologies in tea research and industry. By conducting a detailed analysis of the technical principles of E-nose, GC-MS, and GC-IMS, combined with a comparative examination of typical application cases, this work highlights the unique advantages of each technology in tea-related studies. These cutting-edge research findings not only provide readers with a comprehensive understanding of the technologies but also serve as a methodological reference for practical applications in related fields. Such insights are instrumental in advancing multi-technology integration to drive innovation in tea research and industrial upgrading.

2. Technical principles and current status

2.1. Principles and current status of E-nose

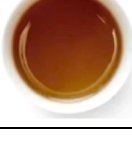
The olfactory system of mammals is not a uniformly organized structure, but rather composed of several subsystems, each potentially serving distinct functions (Breer, Fleischer, & Strotmann, 2006). When external VOCs enter the nasal cavity, they bind to olfactory receptors, generating action potentials. These electrical signals are transmitted along neural pathways to the cerebral cortex, where olfactory perception is formed. Humans possess approximately 400 functional olfactory

receptor types, whose combinatorial interactions enable the recognition of roughly 10,000 distinct odors (Ko & Park, 2016). Sensory analysis is often laborious and costly, particularly when conducted by trained experts who experience olfactory fatigue and cannot sustain prolonged evaluations. This limitation has driven the development of artificial instruments like the E-nose to automate sensory analysis processes.

First proposed in 1982, the E-nose has evolved over four decades (Persaud & Dodd, 1982), mimicking biological olfaction through automated testing cycles (3–5 min per cycle) involving sensor preheating (optimizing headspace saturation), volatile compound sampling, chamber purging, and baseline restoration via fresh air (Sharmilan, Premarathne, Wanniarachchi, Kumari, & Wanniarachchi, 2020). A typical E-nose comprises a multi-sensor array (chemical detection), a processing unit, pattern recognition algorithms, and a reference database. The sensor array mimics the nasal cavity, the processing unit parallels olfactory receptors, and the pattern recognition system with its database emulates cerebral functions for odor identification, enabling systematic analysis of volatile profiles.

Chemical sensors, evolving from early limited tools like ion-selective electrodes, now employ cross-sensitive sensor arrays integrated with pattern recognition (Rudnitskaya & Legin, 2008) to resolve complex mixtures, exemplified by the E-nose, a system whose selectivity, sensitivity, and efficiency depend on sensor types (Gharibzahedi, Barba, Zhou, Wang, & Altintas, 2022). Key sensors operate via electrochemical (redox-driven current/voltage changes), optical (wavelength-specific

Table 1
Comparative analysis table of comprehensive characteristics of six major tea types.

Tea Type	Fermentation Degree	Appearance Characteristics	Liquor Color Characteristics	Liquor Color	Aroma Types	Flavor Profile	TCM Property
Green Tea	0 % (Non-fermented)	Vibrant green, tender leaves; tightly rolled strips	Pale green, bright		Bean, chestnut, fresh	Fresh, mild, and light	Cool
White Tea	5 %-10 % (Slight fermentation)	Plump buds covered with white hairs	Apricot-yellow, translucent		Floral, honey, sweet	Delicate, sweet, and mellow	Cool
Yellow Tea	10 %-20 % (Light fermentation)	Yellowish leaves; uniform strips	Bright yellow		Tender, sweet, toasty	Sweet, smooth, and refreshing	Cool
Oolong Tea	30 %-60 % (Semi-fermented)	Curled leaves; green stems, green belly, red edges	Orange or golden-yellow		Floral, fruity	Rich, mellow, sweet aftertaste	Neutral
Black Tea	80 %-90 % (Full fermentation)	Dark, tightly twisted leaves with golden tips	Deep red, bright		Honey, smoky pine, sweet	Sweet, full-bodied, and velvety	Warm
Dark Tea	Post-fermented	Dark brown, glossy; compressed or loose leaves	Reddish-brown, clear		Earthy, woody, fungal	Mellow, sweet, smooth	Warm

absorption/fluorescence shifts), or piezoelectric (gas adsorption-induced frequency modulation) principles (Chiu & Tang, 2013). Ongoing innovations in sensing materials are advancing multi-component analysis capabilities for complex samples.

E-nose technology plays pivotal roles in food quality control (Jia, Lv, Liu, Zhou, Qin, & Ma, 2024), herbal authentication (Gan, Yang, Gu, Guo, Qiu, Wang, et al., 2024), environmental monitoring (He, Xu, Wang, & Wang, 2017), and medical diagnostics (Dragonieri, Scioscia, Quaranta, Carratu, Venuti, Falcone, et al., 2020). Its future lies in AI integration, where machine learning enables self-learning systems to expand odor databases, refine recognition accuracy for complex profiles, and unlock interdisciplinary applications.

2.2. Principle and current status of GC-MS

GC-MS integrates a gas chromatograph (GC) and a mass spectrometer (MS), where the GC separates chemical mixtures through three core components: (1) the carrier gas system, which uses an inert gas to transport vaporized samples; (2) the injection system, designed to precisely introduce and vaporize controlled sample volumes; and (3) the chromatographic column, which separates compounds based on differences in boiling points, polarity, and adsorption properties. Within the column, components dynamically partition between the stationary phase (column coating) and the mobile phase (carrier gas), with compounds exhibiting higher affinity for the mobile phase (e.g., low-boiling-point or weakly adsorbed substances) eluting first, while those interacting more strongly with the stationary phase elute later, ultimately resolving complex mixtures into individual analytes for downstream analysis.

Prior to GC-MS technology, gas chromatography relied on matching retention times of unknowns to reference standards for qualitative analysis. Under identical conditions (stationary phase, temperature, carrier gas flow), each compound exhibits a specific retention time, allowing tentative identification when unknown peaks align with standards. However, this method faced three critical limitations: (1) Retention times are non-unique, structurally similar isomers often co-elute; (2) Extreme sensitivity to experimental parameters (e.g., minor temperature/flow fluctuations or column batch variations) causes retention time drift, demanding stringent condition control; (3) No structural insights are provided, failing to resolve co-eluting peaks or confirm compound identity, thereby restricting analysis of complex mixtures. Mass spectrometry overcomes traditional qualitative analysis limitations by identifying compounds through mass-to-charge ratio (m/z) measurements. A mass spectrometer comprises three core components: an ion source (e.g., electron impact, chemical ionization, or atmospheric pressure ionization to convert molecules into ions), a mass analyzer (separating ions via electric/magnetic fields, higher m/z ions exhibit larger trajectory curvature), and a detector (registering ion signals) (Alder, Greulich, Kempe, & Vieth, 2006; Epping & Koch, 2023; Fujitake & Harusawa, 2013; Junninen, Ehn, Petäjä, Luosujärvi, Kotiaho, Kostianen, et al., 2010; You et al., 2014). The process generates a mass spectrum (x-axis: m/z ; y-axis: ion intensity), mapping ionized species distribution for structural and compositional analysis.

Since its advent in the 20th century, GC-MS has established itself as a core technology for complex sample analysis, thanks to its high separation efficiency and accurate qualitative capability. This technology has demonstrated irreplaceable value in the food analysis field, with extensive applications and numerous related reports (Table 2).

2.3. Principle and current status of GC-IMS

GC-IMS, as its name implies, is a hyphenated technique combining gas chromatography with ion mobility spectrometry. The gas chromatography component in GC-IMS shares identical principles, components, and functions with that in GC-MS, which we will not elaborate on here. The following discussion focuses on the ion mobility spectrometry

detector, which differs from GC-MS. The IMS primarily consists of three parts: an ionization chamber, a drift tube, and a Faraday plate detector. The ionization chamber serves to convert individual compounds into gas-phase ions using an ionization source (Lv, Mo, Jian, Li, & Guo, 2024). The selection of ionization source significantly affects ionization efficiency, with common β -ionization sources including Nickel-63 (Ni-63), Tritium (H-3), and Americium-241 (Am-241) (Capitain & Weller, 2021). Additionally, atmospheric pressure photoionization, particularly ultraviolet ionization, has emerged as a successful ionization method (Kauppila, Syage, & Benter, 2017). The β -particles emitted by β -ionization sources initiate a series of cascade reactions within the nitrogen environment, primarily generating water cluster protons in the form of $(H_2O)_nH^+$. These ions, typically referred to as "reactant ions," appear as a tall, narrow red peak on the left side of the spectrum known as the Reaction Ion Peak (RIP). When a single compound sample enters the ionization chamber, it combines with reactant ions to form $(H_2O)_nMH^+$. Correspondingly, signal peaks emerge to the right of the RIP peak, representing sample signals. At higher concentrations of single compound samples, dimers may form, and with sufficient concentrations, trimers or even tetramers might develop. However, due to the poor stability of trimers and tetramers under experimental conditions, they are rarely observed in conventional spectra as they tend to decompose or transform during analysis (Pomareda, Guamán, Mohammadnejad, Calvo, Pardo, & Marco, 2012).

Ionization of Drift Nitrogen Gas via β Radiation Source:



Generation of Reactant Ions in Nitrogen:



After ionization, the sample enters the drift tube where ions collide with a counterflow of inert drift gas. Although collisions cause partial velocity loss, the ions regain speed through electric field acceleration, ultimately reaching a constant velocity equilibrium. Due to differences in ion characteristics, such as mass, collision cross-section, charge, and molecular structure, their drift times or drift velocities vary, enabling effective separation even for (Capitain & Weller, 2021; Vautz, Franzke, Zampolli, Elmi, & Liedtke, 2018). However, despite the separation capability of ion mobility spectrometry, peak overlap occurs when compounds exhibit nearly identical drift times and velocities, complicating peak assignment. This necessitates the hyphenation of gas chromatography upstream of IMS to enhance resolution (Aguilera-Herrador, Cárdenas, Ruzsanyi, Sielemann, & Valcárcel, 2008). Finally, ions strike the Faraday detector at distinct times, generating currents upon impact with the metal plate. These currents are converted into signal peaks, completing the sample detection and analysis process.

The equilibrium uniform velocity after collision is referred to as the ion drift velocity (v_d), and the time taken to reach the detector is the drift time (t_d). The product of drift velocity and drift time yields the drift length (L):

$$L = v_d \cdot t_d \quad (7)$$

In the formula: L represents drift length (in cm); v_d represents ion drift velocity (in cm/s); t_d represents drift time (in s).

The ion mobility (K) through an inert gas is related to the weak uniform electric field strength (E) and drift velocity (v_d):

$$K = \frac{v_d}{E} \quad (8)$$

Table 2

Food sample compound analysis research summary table.

Sample Name	Research Category	Chemometric Methods	Column Model	Detected Flavor Compounds	Researchers	Quantitative Indicators and Range
Coconut Jam	Compound Characterization	PCA	ZB-5MS (30 m × 0.25 mm × 0.25 μm)	33 compounds, including esters (ethyl decanoate), aldehydes (hexanal, furfural), ketones (2-octanone), and alcohols (2-octanol)	(H. Zhang, Chen, Wang, Jiao, Chen, Zhong, et al., 2020)	During the concentration of coconut jam, among 33 aroma compounds, the content of furfural increased with time; aldehydes accounted for approximately 29.7 % of total flavor components. Total relative content of esters: accounting for 22.99 %–31.07 % of volatile components during fermentation, with a significant increase. Among 24 soybean genotypes, the crude protein content ranges from 35.63 % to 43.13 %; total phenols range from 1.15 to 1.77 mg/g, and total flavonoids range from 0.68 to 2.13 mg/g.
Chopped Pepper	Volatile Compound Tracking During Fermentation	PCA	DB-Wax (30 m × 0.25 mm × 0.25 μm)	138 compounds, including hydrocarbons (β-phellandrene), esters (ethyl palmitate), alcohols (citronellol), aldehydes (citral), and sulfur compounds (diallyl disulfide)	(J. J. Wang, Wang, Xiao, Liu, Deng, & Zhou, 2019)	In four sweet potato cultivars (Anna, Jieshu95–16, Ayamurasaki, Shuangzai), a total of 68 volatile compounds were identified, with the total content ranging from 21.33 ± 0.55 μg/g (Ayamurasaki) to 24.98 ± 0.21 μg/g (Anna). Among them, Anna had the highest total content, Jieshu95–16 and Ayamurasaki had relatively lower total contents, and the total content of Shuangzai was 23.4 ± 0.4 μg/g.
Glycine max (L.) Merr	Compound Characterization	PCA	Elite-5 capillary column (30 m × 0.25 mm × 0.25 μm)	88 compounds, including phenols (2,6-dimethoxyphenol, 2-methoxy-4-vinylphenol), ketones (1,2-cyclopentanedione, 3,5-dimethoxycetophenone), esters (methyl hexadecanoate, methyl linoleate), and heterocyclic compounds (2H-pyran-2,6-dione, benzofuran)	(Alghamdi, Khan, El-Harty, Ammar, Farooq, & Migdadi, 2018)	Total volatile esters, quantified by GC-MS via the internal standard 3-nonanone: During post-harvest shelf life at 0d, 3d, 6d and 9d, the content showed an "initial increase then decrease" trend—peaking on 3d at ~15 mg/kg fresh weight, which was significantly higher than 0d (~5 mg/kg), 6d (~10 mg/kg) and 9d (~3 mg/kg) as determined by one-way ANOVA with P < 0.05.
Sweet Potato	Compound Characterization	PCA and PLS-DA	HP-5ms(30 m × 0.25 mm × 0.25 μm)	68 compounds, including aldehydes (hexanal, phenethylaldehyde, nonanal), alcohols (1-hexanol, terpinen-4-ol), ketones (trans-β-ionone), and terpenes (camphene, geraniol)	(R. Zhang, Tang, Jiang, Mo, & Wang, 2021)	Individual fatty acids (percentage of total fatty acids): Linoleic acid: 19.73 ± 0.01 % – 95.44 ± 0.01 %; Palmitic acid: 1.57 ± 0.01 % – 31.83 ± 0.01 %; Oleic acid: 2.06 ± 0.01 % – 24.14 ± 0.01 %;
Peel of European pear cultivar 'Starkrimson' during post-harvest ripening	Compound Characterization and Quantitative Analysis	One-way ANOVA; Duncan's test and Student's <i>t</i> -test	DB-5MS capillary column (30 m × 0.25 mm × 0.25 μm)	VOCs are dominated by esters, aldehydes, terpenes, and other fatty acid derivatives — with esters including hexyl acetate and butyl acetate, aldehydes including hexanal and octanal, and terpenes including farnesene	(Zhou, Wang, Sang, Xu, Zhang, Zhang, et al., 2026)	Aroma compounds (μg/kg, mean ± SD): Benzaldehyde: Ham (9799.78 ± 476.94) > Has (2703.87 ± 118.13); Ethyl benzoate: Has (1734.57 ± 80.15) > Ham (760.23 ± 52.49); OAV: 9 compounds with OAV > 1 (e.g.,
Berries of nine red grapevine cultivars	Compound Characterization and Quantitative Analysis	One-way ANOVA and PCA	Supelcowax 10 capillary column (60 m × 0.25 mm × 0.25 μm)	10 fatty acids: Myristic acid, palmitic acid, cis-7-hexadecenoic acid, stearic acid, oleic acid, vaccenic acid, linoleic acid, α-linolenic acid, arachidic acid, behenic acid; linoleic acid is the most abundant	(Faur, Ola, Bunea, Calugar, Bora, Dulf, et al., 2025)	(continued on next page)
Two cherry laurel fruit varieties (Ham and Has, Prunus laurocerasus L.)	Compound Characterization and Quantitative Analysis	One-way ANOVA and PCA	DB-WAX capillary column (60 m × 0.25 mm × 0.4 μm) and HP-88 capillary column (60 m × 0.25 mm × 0.20 μm)	Aroma compounds: Aldehydes, alcohols, terpenes, acids, esters, amines, furan; dominant compounds: benzaldehyde, ethyl benzoate.	(Kesen & Karka, 2025)	

Table 2 (continued)

Sample Name	Research Category	Chemometric Methods	Column Model	Detected Flavor Compounds	Researchers	Quantitative Indicators and Range
						benzaldehyde OAV: Ham 28.00, Has 7.73; ethyl benzoate OAV: Ham 12.67, Has 28.91).

In the formula: K represents ion mobility (in $\text{cm}^2\text{V}^{-1}\text{s}^{-1}$); v_d represents ion drift velocity (in cm/s); E represents electric field strength (in Vcm^{-1})

To compare mobilities under different environmental conditions, ion mobility is typically converted to the standardized mobility under reference conditions (273 K and 760 mmHg), termed the reduced ion mobility (K_0):

$$K_0 = K \left(\frac{273}{T} \right) \left(\frac{P}{760} \right) = \frac{v_d}{E} \left(\frac{273}{T} \right) \left(\frac{P}{760} \right) = \frac{L}{Et_d} \left(\frac{273}{T} \right) \left(\frac{P}{760} \right) \tag{9}$$

In the formula: K represents ion mobility (in $\text{cm}^2\text{V}^{-1}\text{s}^{-1}$); v_d represents ion drift velocity (in cm/s); E represents electric field strength (in Vcm^{-1}); L represents drift length (in cm); t_d represents drift time (in s); P represents drift gas pressure (in torr); T represents drift gas temperature (in K).

Since the 1990s, when portable GC-IMS systems demonstrated their capability to separate VOCs in headspace gases from complex mixtures, the technology has progressively achieved commercialization, with Germany-based G.A.S GmbH (Gesellschaft für Analytische Sensorsysteme) emerging as the industry leader through its three core products: FlavourSpec®, BreathSpec®, and the GC-IMS Environmental Analyzer. The FlavourSpec® specializes in analyzing VOCs in foods and plants (Chai, Huang, Zhang, Wu, Duan, Yu, et al., 2023; Jiang, Duan, Zhao, Liu, Wen, Zeng, et al., 2023; Márquez-Sillero, Cárdenas, Sielemann, & Valcárcel, 2014; J. Yan, Wang, Yang, Zhang, Cao, Zhao, et al., 2024), facilitating the optimization of food flavors and the extraction of bioactive compounds from botanical sources. The BreathSpec® focuses on detecting exhaled breath VOCs for applications in disease diagnosis, drug metabolism monitoring, and forensic toxicology screening (Qian, 2024; Ruszkiewicz, Sanders, O'Brien, Hempel, Reed, Riepe, et al., 2020; Tiele, Wicaksono, Daulton, Ifeakor, Eyre, Clarke, et al., 2020; Tiele, Wicaksono, Kansara, Arasardnam, & Covington, 2019), while the Environmental Analyzer serves industrial emissions monitoring (natural gas, flue gases, biogas), environmental management, and real-time air quality analysis in space stations (Dai, Yang, Gao, Liao, Wang, & Zhang, 2024; Witkiewicz & Wardencki, 2021; L. Zhang, Li, Chen, Wu, Hu, & Yao, 2022; X. M. Zhu, Yang, Gao, Jiao, Zhao, Yu, et al., 2022). Despite facing developmental challenges, breakthroughs in chromatographic column technology have effectively addressed inherent limitations of ion mobility spectrometry (IMS), propelling GC-IMS toward broader applications and enhanced analytical performance.

2.4. Comparison of data processing workflows

E-nose, GC-MS, and GC-IMS exhibit fundamental differences in data generation. GC-MS produces high-dimensional chromatographic peak and mass spectral data. GC-IMS generates two-dimensional contour plots based on retention and drift times. E-nose yields sensor array response data typically simplified to steady-state values with relatively low dimensionality. Despite these disparities, they share significant commonalities in core data processing workflows. (H. Y. Liu, Wadood, Xia, Liu, Guo, Guo, et al., 2023)

All three require data preprocessing steps including baseline correction, normalization, background subtraction BS, and denoising to enhance data quality (Hayes, Greene, O'Donnell, O'Shea, & Fenelon, 2023). Subsequent feature extraction involves techniques such as peak alignment for GC-MS and GC-IMS, and signal feature extraction for

E-nose. During analysis, multivariate statistical and chemometric methods are extensively applied, including PCA, PLS-DA, OPLS-DA, HCA, SDA, BPNN, LDA, SVM, and VIP analysis (Albaradei, Thafar, Alsaedi, Van Neste, Gojobori, Essack, et al., 2021). The unified objectives are sample classification, identification of key variables driving inter-group differences, specific compounds for GC-MS and GC-IMS, critical sensors for E-nose, and extracting meaningful patterns from complex volatile compound datasets (Okada, Afendi, Altaf-Ul-Amin, Takahashi, Nakamura, & Kanaya, 2010).

Their analytical priorities differ distinctly: E-nose relies on clustering methods such as PCA and classification models including LDA, SVM, and BPNN for rapid identification without providing compound-specific information. GC-MS leverages high separation capability and mass spectral library matching Database Matching for in-depth complex system analysis, compound identification, quantification, and VIP-based biomarker screening(Yu, Yao, & Guo, 2021). GC-IMS utilizes its two-dimensional resolution for rapid qualitative analysis, requiring stringent peak matching, drift correction, and migration time calibration during data processing(Biancolillo & Marini, 2018).

3. Applications of E-nose in tea

3.1. Effective authentication of tea geographical origin

A widely circulated Chinese proverb states, “When oranges grow south of the Huai River, they remain oranges; when planted north, they turn into trifoliate oranges,” highlighting the critical influence of geographical origin on plant growth. This principle equally applies to tea, where regional variations in longitude, latitude, and altitude create distinct environmental conditions, including temperature, humidity, moisture, and light, that significantly alter chemical composition. These differences manifest in variations in leaf morphology, quality, flavor, and medicinal properties, even among teas of the same species.

Huang et al. (J. F. Huang, Yan, Yang, & Xu, 2023) applied ATD-GC-MS (Automatic Thermal Desorption-Gas Chromatography--Mass Spectrometry) and E-nose technologies to analyze the compositional profiles and geographical discrimination of black tea samples from three regions. Using ATD-GC-MS, they identified 71 volatile aroma compounds across black teas of different origins. Statistical analysis of data from both ATD-GC-MS and E-nose revealed clear differentiation among the three regional tea groups. However, in the E-nose combined with OPLS-DA, radar fingerprint plots and scatter plots demonstrated closer clustering between samples from Fu'an City and Datian County, indicating notable sensory similarities in black teas from these two regions. This highlights the complementary role of multi-technique integration in resolving subtle regional distinctions while capturing shared flavor characteristics.

3.2. Characteristic changes of VOCs during tea processing

Rolling plays a critical role in black tea production, significantly influencing its chemical composition, quality, and flavor profile. Following withering, the rolling process disrupts the cellular structure of tea leaves. Its primary objectives include reducing leaf size, breaking cells, extracting sap to form a juice coating on leaf particles, and accelerating chemical reactions.

Wang et al. (Q. W. Wang, Shi, Hu, Tang, Zhou, Wang, et al., 2024)

investigated the dynamic changes of volatile compounds during the rolling of Congou black tea using GC-E-nose and GC-MS techniques. PLS-DA analysis of GC-E-nose data revealed distinct clustering patterns across rolling durations: samples at 15 min, 35 min, and 55 min showed significant separation from both unrolled (0 min) and over-rolled (75 min) groups, indicating marked differences in volatile profiles between initial/intermediate rolling stages and prolonged processing. This aligns with findings by Aaqil et al. (Aaqil, Peng, Kamal, Nawaz, Zhang, & Gong, 2023), demonstrating that extended rolling durations intensify liquor color and pigment content (theaflavins, thearubigins, theabrownins). These collective results suggest rolling time modulates amino acid oxidation, catechin polymerization, and flavonol O-glycoside hydrolysis. Notably, the volatile compound divergence between 55 min and 75 min rolling implies a potential qualitative transition point in chemical composition during this critical phase of black tea processing.

3.3. Classification of tea by quality grades

Wang et al. (L. L. Wang, Xie, Wang, Hu, Jiang, Wang, et al., 2024) employed three bionic technologies, E-nose, electronic tongue (E-tongue), and electronic eye (E-eye), to investigate 56 Congou black tea samples categorized into two distinct quality grades. These biomimetic systems collectively revealed the comprehensive quality characteristics of Congou black tea through their unique analytical strengths. The E-nose detected VOCs, identifying 36 distinct VOCs. The E-tongue analyzed taste profiles across seven dimensions: sour, sweet, bitter, salty, umami, and two composite flavor indices. Simultaneously, the E-eye evaluated color attributes based on brightness, red-green chromaticity, and yellow-blue chromaticity. The identified VOCs, in combination with specific taste and color profiles, endowed each tea sample with unique sensory properties, demonstrating the synergistic power of multi-modal bionic analysis in tea quality assessment.

4. Applications of GC-MS in tea

4.1. Characterization of VOCs in tea

Yin et al. (X. Yin, Huang, Huang, Wu, Tong, Liu, et al., 2022) employed GC-MS combined with multivariate analysis to identify volatile compounds in black tea, detecting a total of 88 compounds. Esters, aldehydes, and alcohols constituted the most abundant categories, with 22, 18, and 17 compounds respectively, serving as primary contributors to the floral, fruity, and sweet characteristics of black tea. Geraniol emerged as the most prevalent compound, exhibiting a distinctive rose-like aroma described as sweet floral-rosy citrus. Beyond its aromatic role, geraniol demonstrates pharmacological properties including insecticidal/repellent, antibacterial, antioxidant, anticancer, and anti-inflammatory activities, while being widely utilized as a fragrance chemical in cosmetics and household products (W. Chen & Viljoen, 2010). Phenylethyl alcohol and phenylacetaldehyde ranked next in abundance. Phenylethyl alcohol, an aromatic alcohol with rose-like floral notes, and phenylacetaldehyde, characterized by honey-like sweetness and floral aroma, undergo interconversion in plants via two core pathways: direct decarboxylation and the Ehrlich indirect pathway. This suggests that phenylalanine in tea leaves may generate phenylacetaldehyde through specific enzymes or the Ehrlich pathway, which is subsequently converted to phenylethyl alcohol via reductase activity. During tea processing, oxidative conditions likely inhibit reductase activity, leading to phenylacetaldehyde accumulation, while the fixation process employs high temperatures to inactivate enzymes and preserve phenylethyl alcohol. Additionally, phenylacetaldehyde may be stored as glycosides and released as aromatic compounds during processing (Günther, Schmidt, Gershenzon, & Köllner, 2019; Koeduka, Fujita, Furuta, Suzuki, Tsuge, & Matsui, 2017). Nonanal demonstrated the highest ROAV (Relative Odor Activity Value), characterized by rose-like,

citrus floral, and creamy aromas – findings consistent with artificial sensory evaluations, further confirming the floral and sweet attributes of black tea.

To gain deeper insights into the actual contribution of volatile compounds to the overall tea aroma, it is critical to first clarify a common misconception: In contributing to the primary aroma, it is not the mere presence of a compound or its high concentration that indicates a greater contribution to the odor; instead, it is determined by its Odor Activity Value (OAV) and Relative Odor Activity Value (ROAV) values. Researchers frequently utilize OAV and ROAV analyses to address this nuance.

OAV is defined as the ratio of a compound's concentration (C) in the sample to its odor detection threshold (OT) ($OAV = C/OT$). An OAV greater than 1 typically indicates that the compound has the potential to contribute to the aroma, and higher values suggest a greater potential impact. However, tea often contains numerous aroma-active compounds, making it difficult to intuitively compare their relative importance based solely on OAV. To address this, the concept of ROAV is introduced. Typically, the compound with the highest OAV in the sample serves as the reference benchmark, assigned $ROAV = 100\%$. The ROAV for other compounds is then calculated using the formula $ROAV_i = (OAV_i/OAV_{max}) \times 100\%$. ROAV clearly reflects the relative contribution intensity of each compound to the overall aroma profile.

In the Yin et al. (2022) study, nonanal, by virtue of its high ROAV, was identified as one of the key aroma compounds most critical to the characteristic floral and sweet notes of black tea. This OAV/ROAV-based analytical approach effectively bridges the precise chemical quantitative data obtained from GC-MS with sensory attributes such as floral and sweet, providing a powerful tool for deciphering the chemical essence of complex tea aromas.

4.2. Characteristic changes of VOCs during tea processing

During tea processing, processing parameters-driven dynamic transformations in chemical composition occur. Integrated analytical technologies enable real-time tracking of manufacturing-generated VOCs, which demonstrate stage-specific compositional patterns (types, concentrations, ratios). Quantifying these variations facilitates precise evaluation of process progression, detecting deviations (over-processing or incomplete stages), and supports precision process optimization and standardized quality maintenance through scientifically guided interventions.

During the manufacturing process of *Citri Exocarpium Rubrum* black tea, Zhou et al. (Q. Y. Zhou, Liang, Ling, Gao, & Ling, 2024) systematically VOCs across four critical processing stages (withering, rolling, fermentation, and drying) using HS-SPME and GC-MS techniques. This blended tea combines the mellow character of black tea with the citrus aroma of *Citri Exocarpium Rubrum*, offering both nutritional value and medicinal potential. The study identified 282 volatile compounds, with terpenes (80 species) and esters (53 species) constituting the predominant aromatic foundation. Terpenes emerged as key aroma contributors due to their low olfactory thresholds and aromatic properties. Notably, D-limonene – a representative terpene with citrus fragrance – serves not only as a crucial component in cosmetics and perfumes but has also demonstrated cardiovascular benefits including vasodilation, heart rate reduction, and blood pressure regulation (Rodríguez, Peris, Redondo, Shimada, Costell, Carbonell, et al., 2017; Sharmeen, Mahomoodally, Zengin, & Maggi, 2021). During withering, terpenoids accounted for 21.30 % of VOCs (with D-limonene at merely 0.42 %), while heterocyclic compounds dominated at 21.96 %, producing an overall grassy odor. The rolling stage with *Citri Exocarpium Rubrum* addition triggered a terpene surge to 54.07 %, marked by substantial increases in D-limonene (19.38 %) and γ -terpinene (9.60 %) presumably derived from the citrus material. Concurrent elevation of 137 metabolites (phenylacetaldehyde, trans-nerolidol) established the foundation for floral-fruity aromas. Minimal metabolic differences were observed

between fermentation and rolling stages, with only 8 altered metabolites, terpenoid proportion at 46.09 %, and reduced heterocyclic compounds facilitating the transition to sweet aromas. Drying achieved peak terpenoid content (61.45 %) through further accumulation of D-limonene (22.37 %), γ -terpinene (14.27 %), and β -damascenone, culminating in the characteristic citrus-black tea aroma profile. Additionally, 46 metabolites (decanal, β -cyclocitral) increased while 28 esters (methyl salicylate) decreased, suppressing grassy notes and enhancing floral-fruity dominance. Detected linalool levels are significant, suggesting it may be a signature compound of black tea. Monoterpene biosynthesis and secondary metabolic pathways were identified as key regulatory mechanisms, particularly during the rolling and drying stages for synthesizing limonene and α -terpineol. These findings provide theoretical foundations for optimizing processing techniques in *Citri Exocarpium Rubrum* black tea production.

4.3. Identification of volatile compounds underlying tea aroma profiles

The aroma profiles of green tea primarily include fresh, chestnut-like, tender, and floral notes. Niu et al. (Niu, Ao, Yu, Zhao, & Huang, 2024) employed GC-MS to analyze volatile compounds in tender-aroma and floral-aroma green teas, identifying 43 compounds in total. While the types of compounds in these two aroma categories were broadly similar, their most abundant classes differed: alcohol compounds dominated in tender-aroma green tea, whereas ester compounds were most diverse in floral-aroma green tea. In both roasted green tea types, the combined content of alcohols and esters exceeded 60 %, while aldehydes, ketones, and heterocyclic compounds showed relatively lower concentrations. Linalool emerged as the most abundant compound in both aroma types, with slightly higher levels in floral-aroma tea compared to tender-aroma tea. As previously noted, linalool's floral and fruity characteristics may explain why floral-aroma teas exhibit more pronounced floral notes due to its marginally higher concentration. 3-Hexenyl hexanoate, the second most abundant compound in floral-aroma tea, contributes a fresh, clean scent to green tea (P. Yin, Kong, Liu, Jiang, Sun, Guo, et al., 2022). Certain low-abundance compounds, such as α -cubebene (a sesquiterpene isolated from the medicinal herb *Schisandra chinensis*), exhibited significant differences between the two aroma types, suggesting that minor constituents may play a critical role in distinguishing green tea aroma profiles. The perceived aroma of compounds is influenced not only by its concentration but also by its odor threshold and odor activity value.

4.4. VOCs variation across storage durations

Certain teas, such as Pu-erh, white tea, and dark tea, share aging characteristics with wine, gaining value over extended storage periods. Market trends reflect this phenomenon: aged Pu-erh and white tea command significantly higher prices than their younger counterparts, epitomized by the saying "one-year tea, three-year medicine, seven-year treasure" for white tea. To investigate flavor evolution during storage, advanced analytical technologies can track dynamic changes in VOCs across storage durations.

Zhang et al. (S. W. Zhang, Sun, Wen, Chen, Sun, Lai, et al., 2023) employed GC-MS to analyze black tea from the years 2015, 2017, 2019, and 2021, identifying 109 compounds in total, with 62 exhibiting significant variations. As storage duration increased, 36 compounds associated with floral and fruity aromas showed a declining trend, while 18 compounds characterized by spicy, woody, and sour notes displayed an upward trend. Sweet aroma components included sulfur compounds, lactones, aldehydes, and ketones, with lactones being the most abundant. In this study, α -ionone, β -ionone, β -cyclocitral, and benzaldehyde were identified as primary contributors to sweet aromas. α -Ionone, a common compound in raspberries and a secondary plant metabolite synthesized endogenously, is widely utilized as a fragrant component in cosmetic (shampoos, soaps) and non-cosmetic products (household

detergents and cleaners) across various industries (Aloum, Alefishat, Adem, & Petroianu, 2020). Floral compounds were predominantly alcohols, aldehydes, acids, and esters, with linalool, 2-phenylethanol, 2-heptanol, and phenylacetaldehyde serving as key contributors. Linalool, a typical floral volatile compound, exhibits floral and fruity notes and demonstrates pharmacological properties such as anti-nociceptive, anti-inflammatory, and neuroprotective activities (Hansen, Frandsen, & Fromberg, 2016).

5. Applications of GC-IMS in tea

5.1. Effective authentication of tea geographical origin

Zheng et al. (Zheng, Gan, Zhao, Chen, Zhang, Qiu, et al., 2023) employed GC-IMS to analyze black teas from four regions, identifying 99 VOCs, with aldehydes and alcohols dominating the profiles. Similarly, Yun et al. (Yun, Cui, Zhang, Zhu, Peng, Cai, et al., 2021) conducted GC-MS analyses on 306 samples from 10 regions, confirming aldehydes, alcohols, and ketones as primary VOCs, aligning with Zheng's findings. (Z)-3-hexenol was universally detected across all black tea samples, with its highest concentration observed in CBT (Yichang black tea, Yichang, Hubei, China), establishing it as a characteristic VOC for CBT. Notably, (Z)-3-hexenol is also prominent in the world's four major black teas (Keemun, Assam, Darjeeling, and Ceylon), suggesting its role as a signature compound for black tea globally (Kang, Yan, Zhu, Liu, Lv, Zhang, et al., 2019). YBT (Yingde black tea, Yingde, Guangdong, China) exhibited elevated levels of 14 compounds compared to other regions, including linalool and benzeneacetaldehyde, which serve as its distinctive markers. In DBT (Dian black tea, Yunnan, China), 18 VOCs, particularly acetophenone and ethyl acetate, showed significantly higher concentrations, distinguishing this variety. KBT (Keemun black tea, Qimen, Anhui, China) demonstrated superior abundance in alcohols, ketones, and aldehydes, with 34 VOCs collectively enabling reliable identification of its origin. These region-specific VOC profiles provide a robust framework for differentiating tea origins, such as distinguishing Fu brick tea from black tea variants. Expanding this methodology could revolutionize origin authentication across diverse tea varieties, offering a scientific foundation for traceability technologies in the tea industry.

5.2. Seasonal harvest variations in tea VOCs

Tea growth and harvesting exhibit seasonal variations, with chemical composition fluctuating alongside natural environmental changes. The annual cycle is divided into four seasons, warm spring, hot summer, cool autumn, and cold winter, with spring, summer, autumn, and winter teas named according to their harvest periods. Spring tea is typically harvested between March and May, summer tea from May to August, autumn tea from August to October, and winter tea from October to November.

Liu et al. (H. C. Liu, Xu, Wu, Wen, Yu, An, et al., 2021) employed GC-IMS to analyze black teas harvested in spring, summer, and autumn, identifying 68 volatile compounds, predominantly aldehydes (28 types) and alcohols (18 types). Chemical composition varied significantly across seasons: spring tea showed notably higher γ -butyrolactone levels compared to summer and winter teas, contributing its characteristic creamy aroma. Summer tea exhibited elevated concentrations of butanoic acid, 2-pentylfuran, (E)-2-octenal, and methyl hexanoate. Butanoic acid (butyric acid), a short-chain fatty acid, is recognized for its beneficial roles in gastrointestinal function (Zaleski, Banaszekiewicz, & Walkowiak, 2013). Autumn tea demonstrated the greatest diversity and abundance of volatile compounds, including linalool, limonene, octan-3-ol, butanal, and butyl butyrate. Common high-abundance compounds across all seasons included hexanol and 3-methylbutanal, the latter, frequently detected in black teas, imparting a malt-like aroma (Jin et al., 2024). Volatile compound profiles generally

followed a U-shaped seasonal trend (decline followed by resurgence), confirming the dynamic nature of tea aroma development.

5.3. VOCs under different drying methods in tea

The drying process is critical in tea processing, serving as a key step during both enzyme inactivation (killing-green) and final drying. During killing-green, drying inhibits enzymatic activity to better preserve functional components like polyphenols, amino acids, and caffeine. Moisture content directly impacts microbial growth, elevated water levels promote bacterial, fungal, and microbial proliferation on tea surfaces, accelerating spoilage and reducing shelf life. Final drying optimizes flavor and aroma profiles while ensuring microbial stability through moisture reduction.

Li et al. (N. N. Li, Yao, Ning, Sun, Lin, Zhu, et al., 2024) employed GC-IMS to analyze volatile compounds in green tea under four drying techniques (hot-air drying, vacuum-freeze drying, vacuum-microwave drying, and vacuum-freeze-microwave drying) and two extraction methods ("dry extraction": direct 80°C extraction; "wet extraction": 100°C soaking followed by 80°C extraction). The study identified 43 VOCs, with alcohols being the most diverse category, followed by aldehydes. Dry extraction detected more VOC types than wet extraction, though certain VOC concentrations were higher in wet extraction. This discrepancy arises because 100°C extraction accelerates VOC release while degrading heat-sensitive compounds, resulting in broader VOC diversity but lower concentrations at 80°C dry extraction. Combination analysis revealed that dry extraction with hot-air drying yielded the lowest VOC concentrations, whereas dry extraction paired with vacuum-freeze-microwave drying achieved peak levels for specific compounds, including 2-methyl-3-(methylthio)furan, (*E*)-2-heptenal, 2, 3-butanedione, and linalool oxide. Wet extraction showed minimal VOC variation across drying methods. Despite its advantages, freeze drying consumes 4–10 times more energy and incurs higher costs than hot-air drying. To address this, efficient physical field-assisted freeze drying technologies, such as ultrasound, microwave, and infrared radiation, have been proposed (Yao, Chen, & Fan, 2023). Further studies demonstrate that freezing and microwave technologies enhance volatile retention in green tea: low temperatures reduce molecular motion and volatile loss, while microwaves promote condensation/polymerization reactions that stabilize volatiles within the tea matrix and facilitate additional volatile release, collectively improving retention (Z. X. Wang, Su, Ren, Lv, Ren, Li, et al., 2022).

5.4. Analysis and identification of VOCs in different tea types

Many tea varieties exhibit high visual similarity, often sharing comparable colors and post-drying morphologies, making it challenging for consumers to distinguish them by appearance alone. Unethical vendors may exploit this ambiguity to sell inferior teas as premium products or adulterate teas with non-authentic materials. Authentication of tea varieties can thus rely on identifying unique VOCs as markers of genuineness (Kaldeli, Zakidou, & Paraskevopoulou, 2024).

Guo et al. (L. Guo, Xie, Zhao, Zhang, & Lin, 2024) applied GC-IMS to analyze VOCs in green, white, yellow, and black teas, successfully identifying 104 compounds, with 35 shared among all four types. White tea contained 78 VOCs, exhibiting greater volatile diversity compared to the other three types. Key findings revealed the universal presence of 2-pentanone, 3-methyl-2-butenal, ethyl 2-methylpentanoate, and their dimers across all teas. Notably, (*E*)-2-hexenal, 2-furfuryl methyl sulfide, and 2-hexanol were absent in yellow tea, potentially linked to its unique yellowing process. This specialized procedure involves sealed yellowing and roasting stages, where sealed yellowing – a critical step for quality optimization – entails maintaining tea leaves in a humid (80 %–100 %) and warm (30°C–50°C) sealed environment to drive complex chemical transformations (X. Feng, Yang, Pan, Zhou, Ma, Ou, et al., 2024). The absence of 1-octene in white and black teas may correlate with their

fermentation protocols. The distinct VOC signatures of each tea type enable precise classification and differentiation through compound analysis, providing critical insights for tea authentication, quality assessment, and chemical evolution studies during processing.

6. Applications of three technologies in tea

GC-IMS, GC-MS, and E-nose exhibit distinct advantages and limitations in analyzing food volatile compounds, as summarized in Table 3.

E-nose offers rapid response, simple operation, and cost-effectiveness, enabling quick characterization of overall flavor profiles and differentiation of aroma differences among samples (Shi, Zhang, & Adhikari, 2018). However, it has limited capacity to detect low-volatile or complex odor compounds, with sensor stability and reproducibility easily affected by environmental factors. Its weak qualitative ability hinders identification of specific characteristic compounds, and results are prone to subjective influences (Gou, Bi, Chen, Wu, Fauconnier, & Qiao, 2023).

GC-IMS combines high sensitivity and rapid analysis, requiring minimal sample pretreatment, operating under atmospheric pressure, and being portable; it generates visual fingerprint spectra and is highly sensitive to short-chain molecules (Song, Ma, Chen, Fei, Tao, Wu, et al., 2025). Yet, its existing qualitative database is narrow, limiting identification of some VOCs. It also has inferior separation capability for complex mixtures compared to GC-MS, with resolution affected by drift tube size and complex quantification processes.

GC-MS boasts high sensitivity, selectivity, and separation efficiency, achieving accurate qualitative and quantitative analysis via comprehensive databases like NIST. It is sensitive to long-chain molecules and can identify diverse volatile compounds (W. L. Li, Zi, Xu, Yang, Dong, Qin, et al., 2024). However, it shows insufficient sensitivity to ultra-trace VOCs, and one-dimensional analysis may suffer from compound co-elution (Z. Y. Li, Yang, Zhang, Shi, Yu, Gao, et al., 2025).

These technologies are complementary: E-nose serves preliminary screening and classification; GC-IMS is ideal for rapid on-site detection with trace analysis capabilities, preserving original flavor profiles; and GC-MS provides definitive qualitative/quantitative validation. The sensitivity of GC-IMS compensates for the limitations of GC-MS in ultratrace detection, while the extensive databases of GC-MS enhance the qualitative accuracy of GC-IMS, forming a multidimensional analytical framework (Gou, Bi, Chen, Wu, Fauconnier, & Qiao, 2023; X. L. Yang, Zhang, Yang, & Xie, 2023; J. X. Yin, Wu, Lin, Li, Ding, Han, et al., 2021).

6.1. Characteristic changes of VOCs during tea processing

Tea is divided into six major categories due to differences in processing techniques, and specific steps in these processes shape the unique aroma of each type (Mozumder, Lee, & Hong, 2025). Specifically, tea aroma formation is closely linked to key production stages—withering, fixation, fermentation/piling, and drying—each with distinct effects: Withering activates enzymatic reactions, promoting the hydrolysis of glycosidic aroma precursors and the synthesis of terpenes, while volatilizing low-boiling green grassy notes (Y. J. Wang, Kan, Thompson, Ling, Ho, Li, et al., 2019). Fixation inactivates enzymes at high temperatures to preserve the fresh and brisk taste, and generates heterocyclic aromatic substances such as pyrazines through the Maillard reaction (L. Zhang, Ho, Zhou, Santos, Armstrong, & Granato, 2019). During fermentation/piling, black tea produces theaflavins and a honey-like aroma through enzymatic oxidation, while dark tea decomposes polysaccharides via microbial metabolism to develop an aged aroma (Z. H. Feng, Li, Li, Wang, Zhang, Wan, et al., 2019). The drying process regulates the retention of moisture and aroma substances through pyrolysis, the Maillard reaction, and carotenoid degradation, ultimately shaping the flavor characteristics of the tea (H. J. Wang, Chen, Xu, Zhao, Wang, Liu, et al., 2024).

Table 3

Comparison of Advantages, Limitations, Linear Range, Limit of Detection (LOD), and Limit of Quantitation (LOQ) of E-nose, GC-MS, and GC-IMS Analysis Methods.

Analytical method	Advantages	Limitations	Linear range, LOD and LOQ	Researchers
E-nose	Rapid real-time detection: Completes odor analysis in seconds to minutes, ideal for on-site or online monitoring. High portability: Compact devices suitable for industrial sites, environmental monitoring, or rapid food safety screening. Simple operation: No complex sample pretreatment; operable by non-specialists. Low cost: Maintenance and operational expenses are significantly lower than advanced instruments.	Low sensitivity: Limited detection capability for trace-level substances. Cross-sensitivity: Sensors are prone to environmental interference (temperature, humidity, other gases), requiring frequent calibration. Database dependency: Models require extensive sample training and perform poorly in identifying novel compounds. No compositional resolution: Provides only overall odor profiles without identifying specific compounds.	The linear range, LOD and LOQ of an electronic nose are not fixed values but vary with the target analyte, sample matrix, and sensor technology. The linear relationship is commonly indicated by the goodness-of-fit of quantitative models, with R^2 values typically between 0.7 and 0.95. Standard e-noses based on metal-oxide-semiconductor sensors usually have LOD and LOQ at ppm levels, yet their accuracy is often limited by sensor cross-sensitivity and matrix effects. In contrast, advanced e-nose systems integrated with GC pre-separation significantly improve performance, achieving LOQs at ppb or even pg levels and LODs down to the pg range. These enhanced capabilities are generally validated through low prediction errors in models such as Partial Least Squares regression.	(Lei, Yuan, Li, Zhou, Li, Zeng, et al., 2023) (Cui, Wang, Yang, Wu, & Wang, 2015) (Pulluri & Kumar, 2022) (X. Zhang, Wu, Lin, Du, Tang, Liu, et al., 2021)
GC-MS	High sensitivity and specificity: Detects compounds at ppb to ppt levels, with precise identification via mass spectral libraries. Quantitative capability: Wide linear range supports accurate concentration analysis. Broad applicability: Effective for comprehensive analysis of complex mixtures. Mature standardization: Reliable methods supported by robust databases.	Time-consuming: Single analysis takes tens of minutes to hours, unsuitable for real-time monitoring. Insufficient sensitivity to specific volatility compounds: Lacks sufficient sensitivity to ultra-high volatility or extremely low volatility compounds. Limited discriminatory ability for isomers and similar compounds: Has limited discriminatory ability for isomers and structurally similar compounds. Significant matrix effect affecting sensitivity: Exhibits a significant matrix effect, which impacts the sensitivity.	From the common characteristics observed across a series of studies, GC-MS consistently demonstrates universal advantages in quantitative performance when detecting various target analytes in different sample matrices. Its linear range typically spans from trace to micro levels, and through the application of matrix-matched calibration curves, matrix interferences are effectively eliminated, ensuring excellent linear correlation. The coefficient of determination R^2 is generally greater than 0.995, laying a solid foundation for accurate quantification of target analytes. The LOD and LOQ of GC-MS are also remarkable, achieving trace-level detection capabilities. Typically, LOD values fall within the range of 0.01–10 ng/mL or $\mu\text{g/kg}$, while LOQ values range from 0.03 to -10 ng/mL or $\mu\text{g/kg}$. The low detection limits are attributed to the high-specificity of GC-MS, which reduces background interference, and the enrichment effects of sample preparation techniques.	(Kamal El-Deen & Shimizu, 2021) (Y. Q. Yang, Zhang, Hua, Deng, Jiang, Li, et al., 2020) (Y. Chen, Lopez, Hayward, Park, Wong, Kim, et al., 2016) (Amini & Hashemi, 2018)
GC-IMS	Rapid analysis: Single detection completes in 5–15 min, faster than conventional GC-MS. Moderate portability: Some devices are handheld or portable, ideal for on-site detection. No vacuum system required: Simplified structure with lower maintenance costs than GC-MS. High sensitivity (VOCs): Detects volatile organic compounds at ppb levels.	Limited resolution: Poor differentiation of isomers or structurally similar compounds. Small databases: Commercial spectral libraries remain underdeveloped, relying on user-built databases. Limited quantitative capability: Narrow linear range, suitable for semi-quantitative analysis. Matrix interference: Complex samples may affect ion mobility, necessitating optimized pretreatment.	The linear range of GC-IMS generally spans 1–2 orders of magnitude. During detection, the monomer-dimer transition that occurs during compound ionization can easily lead to nonlinear responses. However, some studies have employed linearization strategies such as reactive ion peak normalization to effectively extend the linear range, thereby improving the stability of quantitative analysis. GC-IMS demonstrates high sensitivity with LODs generally ranging from ppv to ppb; in specific applications, this can extend to the pg level (mass of analyte). The LOQ is typically 3–10 times of the LOD. Both values are variable and influenced by sample pretreatment, compound properties, and instrument settings.	(Allers, Langejuergen, Gaida, Holz, Schuchardt, Hohlfeld, et al., 2016) (Papp, Nemeth, Siyapndjeu, Bufa, Marosvölgyi, & Gyöngyi, 2024) (Schanzmann, Gaar, Keip, Telgheder, & Sielemann, 2025) (Thomas, Zeh, Dörfel, Zhang, & Hinrichs, 2021)

Yang et al. (Y. Q. Yang, Wang, Xie, Deng, Zhu, Xie, et al., 2024) employed GC-E-Nose, GC-IMS, and GC-MS to investigate VOCs across the entire processing chain of sweet-floral black tea (fresh leaves $\rightarrow$ withering $\rightarrow$ rolling $\rightarrow$ fermentation $\rightarrow$ drying $\rightarrow$ firing). GC-E-Nose revealed continuous VOC fluctuations, particularly significant during shaking and rolling stages. GC-IMS detected 82 VOCs and GC-MS identified 106 VOCs, predominantly alcohols, alkenes, aldehydes, and esters. Fresh leaves showed higher levels of 1-pentanol, benzaldehyde, and ethyl sulfide than any processed stage. Notably, 1-pentanol contributes to the green/grassy-roast aroma in black tea (F. Yan, Chen, Qu, Huang, He, Wan, et al., 2024), while benzaldehyde, identified as a key almond-like aroma marker in wild black tea by Wang et al. (Q. S. Wang, Jiang, Qin, Liu, Li, Fang, et al., 2020), serves as an indicator for almond-scented varieties. During withering, compounds like

2-heptanone, 2-methyl-1-propanol, hexanol, 2-methylbutane-1-ol, methyl valerate, butyric acid, and ethyl 2-ethyl-3-methylbutyrate peaked. Rolling significantly increased methyl acetate, butanal, octanal, cis-linalool oxide, and trans-linalool oxide. Fermentation elevated concentrations of tert-butanol, (*E*)-2-heptenal, 2-acetylfuran, camphene, and ethyl 3-hydroxybutyrate. Microbial-driven fermentation initiates enzymatic and non-enzymatic reactions, particularly glycoside hydrolysis, which critically shapes the tea's aromatic profile by generating alcohols and aldehydes (Hu, Shi, & Ma, 2022). Drying enriched aldehyde compounds such as pentanal, heptanal, nonanal, and hexanal, while firing increased (*E*)-3-penten-2-one, 3-hydroxybutan-2-one, and furfural.

In a parallel study on chestnut-aroma green tea processing (fresh leaves $\rightarrow$ withering $\rightarrow$ enzyme inactivation $\rightarrow$ rolling $\rightarrow$ drying; omitting

fermentation due to non-oxidative nature), Yang et al. (Yang et al., 2022) observed dynamic VOC variations across stages using the same technologies. E-nose detected maximal VOC changes during drying, which promoted Maillard reaction-derived heterocyclic compounds (pyrrole, methylpyrazine) and sulfur-containing substances essential for chestnut aroma formation. GC-IMS identified 33 VOCs: methyl benzoate, methyl hexanoate, butyl acetate-D, ethyl propionate, and isopropyl acetate dominated fresh leaves; alcohols (ethanol, hexanol, butanol) peaked during withering; (Z)-3-hexen-1-ol-D increased during enzyme inactivation, a thermal process that deactivates enzymes to preserve green tea's "clear liquor and green leaves" quality by facilitating targeted biochemical transformations (Y. C. Li, Ni, Jiang, Zhang, Zhou, Li, et al., 2023). Rolling elevated heptanal-M, hexanal, and pentanal, likely influenced by cellular disruption that releases polyphenols (catechins) from chloroplasts and vacuoles, potentially inhibiting terpene alcohol formation (S. Chen, Liu, Zhao, Li, Shan, Wang, et al., 2020). Drying enriched nonanal, octanal-M, and (E,E)-2,4-heptadienal. GC-MS detected 211 compounds, with VOC counts per stage as follows: fresh leaves (174 types), withering (170 types), enzyme inactivation (172 types), rolling (162 types), and drying (202). Alcohols and aldehydes peaked during withering, while alkanes and alkenes dominated fresh leaves. Esters, terpenes, and organic acids declined throughout processing, whereas heterocyclic compounds showed an initial decrease followed by resurgence.

6.2. VOCs dynamics during tea fermentation

Tea fermentation significantly enhances flavor and taste by oxidizing tea polyphenols, which dominate the bitterness and astringency of unfermented leaves. During fermentation, polyphenol oxidation reduces their concentration, mitigating harshness and creating mellower flavors. Concurrently, the aroma profile evolves from fresh grassy notes to complex fruity and floral bouquets (Assad, Ashaolu, Khalifa, Baky, & Farag, 2023; Ghosh, Bag, Sharma, Tudu, Sabhapondit, Baruah, et al., 2015). Microbial and enzymatic activity during fermentation also breaks down macromolecules into smaller, bioavailable compounds, improving nutrient absorption and health benefits (Hu, Shi, & Ma, 2022).

Xiao et al. (Xiao, Huang, Chen, Zhu, He, Li, et al., 2022) investigated VOC dynamics in E. cristatum-fermented Fu brick tea using HS-GC-IMS, HS-SPME-GC-MS, and E-nose. The fungus exerted maximal flavor-modifying effects during mid-fermentation. HS-GC-IMS identified 54 VOCs, predominantly aldehydes (21 types), esters (9), and alcohols (7). Aldehydes (pentanal, hexanal) increased from 42.58 % to 51.22 %, while alcohols (ethanol, benzyl alcohol) generally rose, except for 1-octen-3-ol, which dipped mid-fermentation before rebounding. Esters (ethyl benzoate, methyl 2-methylbutyrate) accumulated, though ethyl acetate declined. Mid-fermentation fingerprinting revealed unique VOC disappearance/appearance patterns, marking this phase as critical for flavor development. HS-SPME-GC-MS detected 93 VOCs, with ketones and hydrocarbons predominating. Hydrocarbon accumulation aligned with fermentation progression. E-nose and PCA confirmed stark aroma differences between unfermented and fermented teas, particularly post-mid-fermentation.

Zhang et al. (D. Zhang, Huang, Fan, & Zeng, 2024) analyzed A. cristatus-fermented Fu brick tea sampled every two days (days 0–14). E-nose detected intensified responses to nitrogen compounds (W5S), short-chain alkanes (W1S), aldehydes/ketones (W2S), and aromatics (W2W), indicating VOC enrichment. HS-SPME-GC-MS identified 112 VOCs, with ketones (24 types), alcohols (22 types), and hydrocarbons (22 types) dominating. VOC diversity peaked at day 6 before declining. Ketones (6-methyl-5-hepten-2-one, α -ionone) and alcohols (linalool, cis-verbenol) increased significantly, with alcohol content rising from 5.37 % to 25.61 %. GC-IMS revealed 76 compounds, showing aldehyde reduction (hexanal, 2-hexenal) and late-emerging compounds (2, 3-butanedione, 1-octen-3-ol). Some VOCs (2-octanone, 2,6-dimethylpyrazine) exhibited bell-shaped concentration curves.

6.3. Analysis and identification of VOCs in different tea types

Liu et al. (Z. Liu, Ma, Liu, Dong, Wang, Zhang, et al., 2024) investigated VOCs in different rose tea varieties (Rosa spp.) using GC-IMS and GC-MS. GC-IMS identified 15 potentially discriminatory compounds, with 1-penten-3-one and ethyl propionate showing the most significant variation across six rose cultivars. GC-MS revealed distinct compositional profiles: XS (Rosa centifolia "Xiangshui") exhibited higher aldehydes, alcohols, alkanes, and phenols, while KS (Rosa sertata $\times$ Rosa rugosa cv. Plena) was rich in ketones, heterocyclics, and ethers. Phenylethyl alcohol, citronellol, ethyl octanoate, and ethyl decanoate emerged as key aroma contributors. Phenylethyl alcohol, a phenylalanine-derived volatile (Hua & Xu, 2011), imparts a classic rose fragrance, validated in multiple studies on odor perception. Citronellol, a precursor to rose oxide (a monoterpene critical in perfumery), plays a pivotal role in shaping rose aroma complexity (Dwivedi, Singh, Singh, Jatav, Sangwan, & Mishra, 2018; Ren, Xue, Sun, & Zhao, 2018).

P. Ginseng (PGF), P. quinquefolius (PQF), and P. notoginseng (PNF) are commonly used in Chinese herbal teas but are frequently misidentified due to visual and taxonomic similarities. Yue et al. (Yue, Yin, Xie, Wu, Ding, Han, et al., 2024) employed GC-E-Nose, GC-IMS, and GC-MS to differentiate these species via floral bud VOCs. GC-E-Nose revealed distinct profiles: PNF buds emitted 26 VOCs dominated by fruity/grassy notes ((Z)-3-hexenal, pentadecane), while PQF (11 VOCs) and PGF (14 VOCs) shared cocoa-spicy aromas with elevated chloroethane and esters. GC-IMS identified 68 VOCs: PNF contained 58 (β -ocimene, α -farnesene, D-limonene), PQF 48 (α -terpinene, phenylacetaldehyde), and PGF 42 (α -phellandrene, α -pinene). GC-MS detected 78 VOCs, with PNF characterized by octanal, β -elemene, and germacrene D; PGF and PQF shared β -elemene and santalene but diverged in sesquiterpene profiles. These findings establish VOC fingerprints for precise species authentication.

6.4. VOCs variation across storage durations

Rong et al. (Rong, Xie, Yuan, Wang, Liu, Deng, et al., 2023) applied GC-E-Nose, GC-MS, and GC-IMS to analyze VOCs in Pu-erh tea stored for 1–4 years (S1Y–S4Y). GC-E-Nose revealed progressive VOC profile shifts, with S4Y diverging notably from earlier years, suggesting the third to fourth year marks a critical transition. GC-MS identified 43 VOCs dominated by aldehydes, ketones, alcohols, and heterocyclic compounds. S2Y exhibited peak levels of aldehydes, alcohols, esters, and hydrocarbons, while S3Y showed minimal concentrations. Heterocyclic and ketonic compounds peaked in S1Y but plummeted in S3Y. Statistical analysis highlighted significant compositional similarities between S3Y and S4Y, contrasting sharply with S2Y, indicating fundamental VOC transformations during the third year. GC-IMS detected 91 VOCs, with aldehydes, alcohols, and ketones predominating. Storage year-specific patterns emerged: S1Y featured elevated heterocyclics (pyrazines, methylthiophene) and esters; S2Y accumulated terpenic alkenes (α -pinene, camphene); S3Y shifted toward ketones (6-methyl-5-hepten-2-one) and alcohols; S4Y prioritized aldehydes (nonanal, heptanal). These distinct VOC signatures enable reliable age authentication for Pu-erh.

Huang et al. (H. R. Huang, Chen, Wang, Cheng, Liu, Hu, et al., 2024) employed E-nose, HS-GC-IMS, and HS-SPME-GC-MS to profile VOCs in white teas stored from 2020 to 2023. E-nose response patterns indicated high inorganic/organic sulfides and aromatics but low alkanes and ammonia. HS-GC-IMS identified 55 VOCs, with aldehydes (29.9 % $\rightarrow$ 33.4 %), acids (6.8 % $\rightarrow$ 9.4 %), and fluctuating alcohols (24.2 % $\rightarrow$ 25.4 % $\rightarrow$ 23.3 %) dominating. Ketones declined sharply (11.9 % $\rightarrow$ 7.0 %). Compound-specific trends emerged: 2-octanol, linalool, and ethyl acetate decreased linearly; 2-heptanol and borneol acetate peaked mid-storage; acetic acid and 1-butanol increased steadily. Linalool decreased while linalool oxide rose, aligning with Zhu et al. (Y. Zhu, Kang, Yan, Lv, Zhang, & Lin, 2020). HS-SPME-GC-MS detected 53 VOCs,

predominantly hydrocarbons, alcohols, and esters. Alcohols, esters, and ketones declined over time, aldehydes peaked mid-storage, and acids rose continuously. PCA revealed compositional convergence between second- and third-year teas, identifying year three as a pivotal transition, while first- and fourth-year teas diverged markedly, confirming storage-driven flavor metamorphosis.

7. Summary and Prospects

The integration of E-nose, GC-MS, and GC-IMS has exhibited substantial potential in advancing the analysis of VOCs in tea.

The E-nose, characterized by its rapid detection and real-time monitoring capabilities, functions as a pragmatic tool for preliminary screening and classification of tea quality and origin. However, its limitations in sensitivity and inability to identify specific compounds necessitate supplementation with other analytical techniques.

As the gold standard for compound identification, GC-MS provides precise structural elucidation and quantification of VOCs. Importantly, GC-MS exhibits high sensitivity in detecting long-chain volatile compounds, whereas GC-IMS offers superior sensitivity and detection performance for short-chain volatile molecules—thus forming a complementary analytical spectrum.

Furthermore, GC-IMS imposes lower requirements for sample pretreatment and operates under atmospheric pressure, which simplifies experimental procedures and reduces operational complexity. Nevertheless, GC-IMS has its inherent limitations: due to its relatively shorter development history compared to GC-MS, the available compound databases for GC-IMS are less comprehensive. This deficiency in database coverage poses challenges for qualitative analysis, as many compounds may be absent from reference libraries, leading to potential uncertainties in compound identification.

The synergistic application of these three detection and analysis technologies is what can better drive the sustainable development of the tea industry. Tea has gained global popularity, with its captivating aroma being a core factor underlying this appeal, and these three technologies enable precise identification of tea aroma components. Historically, the development of the tea industry relied heavily on practitioners' experience, lacking scientific standardization. However, the standardized application of these technologies now facilitates the transformation of tea aroma quality control from an experience-based model to a data-driven paradigm, which in turn supports the standardized production of high-quality tea products and bolsters brand growth.

Future research should prioritize the integration of artificial intelligence (AI) with E-nose, GC-MS, and GC-IMS, leveraging machine learning algorithms to optimize data fusion. For instance, the rapid fingerprinting of the E-nose can be coupled with GC-MS/GC-IMS spectral databases to enhance its compound identification capability, while AI-modeled correlations with GC-MS data can improve the quantification accuracy of GC-IMS. Additionally, developing a miniaturized, integrated system that synergizes the three technologies could address their respective limitations: it would incorporate the real-time responsiveness of the E-nose, the high-resolution compound identification of GC-MS, and the on-site sensitivity of GC-IMS, while mitigating the cross-sensitivity of the E-nose, the lengthy analysis time of GC-MS, and the database constraints of GC-IMS. Such advancements would enable seamless, high-throughput analysis of tea VOCs across production chains, from rapid on-site screening to precise molecular characterization, ultimately driving smarter quality control and innovation in tea science.

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CRediT authorship contribution statement

Zhongyu Li: Writing – original draft, Investigation, Data curation. **Jianya Ling:** Writing – review & editing, Supervision. **Huaijie Shi:** Data curation. **Jiixin Yu:** Investigation. **Zhaolong Gao:** Data curation. **Guoying Zhang:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Data availability

No data was used for the research described in the article.

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